Lecture 4.1: Conceptual Framework and Theory

AREC 570 / ECON 530 · Unit 4

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Readings and sequence

1. **Ethridge, Chapter 8**
    - What a conceptual framework does in applied economic research
2. **Varian, “How to Build an Economic Model in Your Spare Time”**
    - How to isolate a mechanism and structure a useful model
3. **Romer, “Mathiness in the Theory of Economic Growth”**
    - How to keep verbal mathematical and empirical arguments aligned

Example: SNAP retailer stocking requirements

SNAP-authorized retailers must satisfy program eligibility standards. Eligibility requires a minimum assortment of staple foods.

In 2018, a new rule went into effect that changed the minimum number of items per food category to remain eligible.

Changes in how qualifying varieties (breadth) and stocking units (depth) are defined may affect

- Products carried by participating retailers
- Cost of remaining SNAP authorized
- Retailer entry into the program *and/or* continued participation *and/or* exit
- Access to staple foods for SNAP households

The core policy issue and rationale

Policy concern

Previous minimum standards could allow SNAP participation with limited nutritious variety

Policy intervention

Require more varieties across staple food categories and more perishable foods

Intended mechanism

Use the value of SNAP authorization to encourage neighborhood retailers to stock more nutritious foods

Longer-run goal

Improve convenient access to foods consistent with dietary guidance in low-income and rural communities

Translating the policy issue into a research problem

Intended stocking pathway

stocking requirement → retailer inventory → local availability → purchases and health

Retailer participation pathway

stocking requirement → compliance cost → SNAP authorization → authorized-store portfoli

For purposes of this example, the application observes the *second* pathway.

Begin with a researchable problem

Broad policy question

> Do stronger stocking standards improve access to staple foods?

Researchable questions

1. How do revised standards affect continued SNAP authorization?
2. Which types of retailers are more likely to lose authorization?
3. How does the portfolio and location of authorized retailers change?
4. Are authorization losses concentrated in areas with a higher share of SNAP-participating households?

Where does the conceptual framework enter?

These parts often develop together while writing a manuscript

- The **problem statement** identifies the possible loss of SNAP-authorized retailers
- The **literature review** offers theory and evidence about regulation retailer participation and food access
- The **conceptual framework** explains why the rule may change authorization and predicts which stores and communities are most affected
- The **methods** use annual store authorization and location data to evaluate those predictions

Distinguish the conceptual framework from the empirical design

- The framework states the expected relationships
- The empirical design specifies how the evidence will be used to estimate them

Ethridge, Chapter 8

Ethridge: conceptualize the problem

For Ethridge, a conceptual framework is an analysis of a specific research problem using relevant theory and concepts (Ethridge 2004).

The framework applies theory to explain the relationships that create or characterize the problem.

It is not a review of standard theory and it is not a description of the empirical method.

Five elements of a conceptual framework

Ethridge draws on Williams to describe an organized analysis that may include

- Sources of the problem
- Proposed solutions to the problem
- Variables relevant to the analysis
- Relationships organized into a system
- Hypotheses about expected results

Source: Ethridge Chapter 8 pages 130–131

Applying the five elements to the SNAP example

Source of the problem Some SNAP retailers may offer limited nutritious variety

Proposed solution Stronger stocking standards

Relevant variables Authorization status, store type, location, and neighborhood-level SNAP participation rates

Relationship Compliance costs alter the retailer's decision to remain authorized

Expected result Authorization losses vary across store types and communities

Three kinds of hypotheses in Ethridge

Type	Role in the conceptual framework
Maintained hypotheses	Assumptions accepted for purposes of the analysis
Diagnostic hypotheses	Proposed explanations for the source or nature of the problem
Remedial hypotheses	Proposed actions expected to address the diagnosed problem

Maintained hypotheses define the conditions for the reasoning. Diagnostic hypotheses explain the problem. Remedial hypotheses connect a proposed intervention to a response.

Apply the three hypothesis types

Maintained hypothesis

Meeting the stronger stocking standard is costly for at least some retailers

Diagnostic hypothesis

Some SNAP households have limited access to nutritious foods because nearby retailers stock too few qualifying products

Remedial hypothesis

A stronger stocking standard increases nutritious inventory among retailers that comply and remain authorized

The conceptual framework also identifies competing responses

The intended policy mechanism is not the only possible response.

- Some retailers already meet the new standard
- Some retailers change inventory and remain authorized
- Some retailers decide that compliance is not worthwhile
- Some retailers lose authorization
- Authorization losses may be concentrated by store type or community

The framework distinguishes the unobserved stocking mechanism from the observed authorization outcome.

How Ethridge recommends proceeding

1. Review how related problems have been conceptualized
2. Reduce the problem to its simplest useful form
3. Identify the applicable economic theory
4. Construct a base model using maintained hypotheses
5. Add dimensions only when they improve the analysis
6. Assemble hypotheses from the conceptual reasoning

Source: Ethridge Chapter 8 pages 136–138

Discussion: apply Ethridge to your research

1. What specific condition or relationship requires explanation in your project
2. Which theory helps you reason through that problem
3. What maintained hypothesis does your reasoning require
4. What diagnostic hypothesis follows from the framework
5. What remedial hypothesis would correspond to a proposed intervention

Varian

From the concept to the model

Ethridge asks what the conceptual framework must accomplish

Apply relevant theory to the research problem and derive meaningful hypotheses

Varian asks how the economic model should be structured

Identify agents, choices, objectives, constraints, interactions, and adjustment

Varian: start with the following questions

1. Who makes choices?
2. What can each agent choose?
3. What objective guides each choice?
4. What constraints limit each agent?
5. How do the agents interact?
6. What adjusts when choices are not mutually consistent?

Work through the simplest example that retains the mechanism. Add complexity only when it changes the result (Varian 1997).

Structure the example: agents and choices

Who makes choices and what can they choose?

Retailers decide whether the value of SNAP authorization justifies the cost of meeting the rule

Households choose among the retailers available to them

The regulator sets stocking standards and determines eligibility

Structure the example: objectives and constraints

What do agents want and what limits their choices?

Retailers value SNAP-related revenue and face compliance procurement shelf-space and spoilage costs

Households value food and other goods and face income travel time prices and product availability

The regulator seeks healthier food access and program integrity within administrative constraints

Structure the example: interaction and adjustment

How do choices interact and what adjusts?

The stocking standard changes compliance costs for some retailers

The value of SNAP customers affects a retailer’s willingness to incur those costs

Retailers enter remain in or leave the authorized-store network

The number type and location of authorized stores adjust across communities

A first model: retailer authorization

Let r represent the strictness of the stocking standard.

For retailer i

$$B_i = \text{value of SNAP authorization}$$
$$K_i(r) = \text{cost of meeting the stocking standard}$$

The observed authorization outcome is

$$a_i(r) = \begin{cases} 1 & \text{if } B_i \geq K_i(r) \\ 0 & \text{if } B_i < K_i(r) \end{cases}$$

When does the rule force a change?

Requires a change — binding

The new standard requires a retailer to change its inventory to remain eligible

Already satisfied — slack

The retailer already meets the standard so the rule does not require an inventory change

Inventory without the program minimum — unconstrained inventory

The inventory a retailer would choose without the program minimum. It does not mean that the retailer is exempt from the rule.

Developing a conceptual model

For market or neighborhood m the number of authorized retailers is

$$A_m(r) = \sum_{i \in m} a_i(r)$$

The portfolio also describes

- The number of authorized stores
- The types of stores that remain
- Their geographic distribution
- The communities that experience losses

Here $A_m(r)$ is the number of authorized retailers in market m under stocking standard r. The outcome is the authorized-store network rather than the quantity of inventory.

Program targeting

Let S_m denote the share of households participating in SNAP in community m.

If authorization losses are larger where S_m is higher then the authorized-store network contracts most in communities that rely more heavily on SNAP.

That pattern would suggest a targeting problem relative to the policy's food-access goal.

It would not by itself establish the effect on store inventory household purchases diet or health.

From the model to observable measures

Model object	Empirical counterpart
r	Implementation of the revised stocking standard
a_{it}	Store i authorization status in year t
A_{mt}	Number and composition of authorized stores in market m and year t
S_{mt}	Share of households participating in SNAP

The empirical analysis asks how a_{it} and A_{mt} changed after the policy and whether those changes vary with S_{mt}.

Maintained hypotheses behind the result

- Meeting the new standard is more costly for some retailers than for others
- SNAP authorization has positive but heterogeneous value across retailers
- Retailers can remain authorized lose authorization or operate outside SNAP
- The policy can change the authorized-store portfolio even when inventory is not observed

These maintained hypotheses support the authorization mechanism. They do not establish its empirical magnitude.

Match the model to available evidence

Available data	Outcome supported by the data
Store existence by year	Entry survival and exit
SNAP authorization by year	Entry continuation and loss of authorization
Store characteristics	Composition of the authorized-store portfolio
Store location	Geographic distribution of authorized retailers
Local SNAP participation	Distribution of authorization losses by program reliance

Mechanisms outside the available data

- Inventory changes explain how stores may comply
- Compliance costs explain why authorization responses may differ
- SNAP revenue explains why authorization is more valuable to some stores
- Household purchases diet and health lie farther along the policy pathway

These mechanisms motivate the framework but are not outcomes that this application can test directly.

Discussion: apply Varian to your research

1. Who makes the central choice in your model
2. What is the smallest set of choices objectives and constraints needed for the mechanism
3. What adjusts when agents make incompatible choices
4. Which feature can be removed without changing the prediction
5. Which extension could change the prediction

Romer

From the model to the argument

Varian provides the structure of the economic model

Agents, choices, objectives, constraints, interactions, and adjustment

Romer asks whether the model supports the claims made for it

Words symbols and empirical interpretations must remain connected

Romer: keep words and symbols connected

Romer's concern is not the use of mathematics. It is slippage among

- What the verbal argument says
- What the mathematical model establishes
- What the empirical evidence can support

Mathematical notation improves the argument only when terms are defined consistently and the conclusion follows from the stated assumptions (Romer 2015).

Audit the claim

Claim

> Stricter stocking requirements improve food access

Claim supported by the available outcomes

> Revised stocking standards may change the number type and location of SNAP-authorized retailers

If authorization losses are concentrated in high-SNAP communities then the pattern is inconsistent with the policy’s targeting objective.

What the available data do not establish

The available data do not establish

- How store inventory changed
- Whether remaining stores offer more nutritious foods
- How household purchases changed
- Whether diet or health improved

A Romer-style consistency check

Verbal statement	Mathematical statement	Evidence required
Compliance affects authorization	$a_i(r)$ depends on B_i and $K_i(r)$	Store-year authorization histories
The authorized portfolio changes	$A_{mt} = \sum_{i \in m} a_{it}$	Store type location and authorization
Losses are concentrated in high-SNAP areas	Authorization losses rise with S_{mt}	Local authorization changes and SNAP participation
Nutritious inventory improves	Inventory rises among continuing stores	Product-level inventory or audit data

If a row cannot be completed coherently then the claim or model needs revision.

Discussion: apply Romer to your research

1. Which term in your argument is most likely to change meaning
2. Which claim is theoretical and which claim is empirical
3. Does each symbol have a stable verbal interpretation
4. What conclusion does the model support
5. What stronger conclusion would require additional theory or evidence

Synthesis

Summary of readings

Reading	Central question	Contribution to a research project
Ethridge	What should the conceptual framework accomplish	Defines the problem relationships and hypotheses
Varian	How should the economic model be structured	Identifies agents, choices, objectives, constraints, interactions, and adjustment
Romer	How should the argument be disciplined	Keeps words symbols and empirical claims aligned

Together the readings turn a broad assertion into conditional and refutable claims.

What belongs in the written framework?

For your project the conceptual framework should state

1. The problem that requires explanation
2. The relevant theory and concepts
3. The agents, choices, objectives, and constraints
4. The mechanisms connecting the proposed cause to the outcome
5. The maintained diagnostic and remedial hypotheses
6. The conditions under which each prediction applies

Research-design implications

The framework generates design questions

- What is the appropriate unit of analysis
- How will loss of authorization be distinguished from store closure
- Which store characteristics define the portfolio
- Which geography defines the relevant community
- How will local SNAP participation be measured
- When can each outcome respond
- Which adjustments could affect the comparison group

These questions should inform the empirical design.

Apply the sequence to your research

1. **Problem** State the condition that requires explanation
2. **Theory** Identify the concepts that help explain it
3. **Structure** Identify agents, choices, objectives, constraints, and adjustment
4. **Prediction** State what should change and under which conditions
5. **Boundary** State what the model does not establish
6. **Evidence** Identify an observation that would challenge the proposed mechanism

References

Ethridge, Don E. 2004. “The Conceptual Framework.” Chap. 8 in *Research Methodology in Applied Economics: Organizing, Planning, and Conducting Economic Research*, 2nd ed. Blackwell Publishing.

Romer, Paul M. 2015. “Mathiness in the Theory of Economic Growth.” *American Economic Review: Papers & Proceedings* 105 (5): 89–93. https://doi.org/10.1257/aer.p20151066.

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